



Evaluation Metrics

Regression & Classification

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Metrics

Outline

Regression Metrics

Classification
Metrics

Outline

- 1 Regression Metrics
- 2 Classification Metrics



Metrics

1 Outline

Regression Metrics

Classification
Metrics



Metrics

Outline

2

Regression Metrics

Classification
Metrics

Regression Metrics

Regression Metrics

Problem Setting and Notation

Regression Problem

Given input $\mathbf{x}$, predict a real-valued target $y \in \mathbb{R}$.

$$\hat{y} = f(\mathbf{x})$$

Residual / Error

For each sample i :

$$e_i = y_i - \hat{y}_i$$

Key Idea

Regression metrics summarize the magnitude and structure of prediction errors.



Metrics

Outline

3

Regression Metrics

Classification
Metrics

Regression Metrics

MAE, MSE, RMSE

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Squared Error (MSE) and Root MSE (RMSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}$$

Interpretation

- MAE: more robust to outliers (linear penalty).
- MSE/RMSE: penalize large errors more (quadratic penalty).

Metrics

Outline

4 Regression Metrics

Classification
Metrics

Regression Metrics

Step-by-step Example: MAE / MSE / RMSE

Example Data (n=4)

$$\mathbf{y} = [3, -0.5, 2, 7], \quad \hat{\mathbf{y}} = [2.5, 0, 2, 8]$$

i	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$	$ e_i $
1	3.0	2.5	0.5	0.5
2	-0.5	0.0	-0.5	0.5
3	2.0	2.0	0.0	0.0
4	7.0	8.0	-1.0	1.0
			Sum	2.0

$$\text{MAE} = \frac{2.0}{4} = 0.5$$

Metrics

Outline

5

Regression Metrics

Classification
Metrics

Regression Metrics

Step-by-step Example: MSE / RMSE

Using the same example:

$$\mathbf{e} = [0.5, -0.5, 0, -1]$$

i	e_i	e_i^2
1	0.5	0.25
2	-0.5	0.25
3	0.0	0.00
4	-1.0	1.00
Sum		1.50

$$\text{MSE} = \frac{1.50}{4} = 0.375, \quad \text{RMSE} = \sqrt{0.375} \approx 0.612$$

Metrics

Outline

6

Regression Metrics

Classification
Metrics

Takeaway

Regression Metrics

R^2 (Coefficient of Determination)

Definition

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Interpretation

- $R^2 = 1$: perfect prediction.
- $R^2 = 0$: same as predicting the mean $\bar{y}$.
- $R^2 < 0$: worse than predicting the mean.

Metrics

Outline

7

Regression Metrics

Classification
Metrics

Regression Metrics

Step-by-step Example: R^2

Example:

$$\mathbf{y} = [3, -0.5, 2, 7], \quad \hat{\mathbf{y}} = [2.5, 0, 2, 8]$$

$$\bar{y} = \frac{3 - 0.5 + 2 + 7}{4} = \frac{11.5}{4} = 2.875$$

Compute SSE and SST

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2 = 1.50$$

$$\text{SST} = \sum (y_i - \bar{y})^2$$

Now compute SST:

$$\begin{aligned} & (3 - 2.875)^2 + (-0.5 - 2.875)^2 + (2 - 2.875)^2 + (7 - 2.875)^2 \\ &= (0.125)^2 + (-3.375)^2 + (-0.875)^2 + (4.125)^2 \\ &= 0.015625 + 11.390625 + 0.765625 + 17.015625 = 29.1875 \end{aligned}$$

Metrics

Outline

8

Regression Metrics

Classification
Metrics

Regression Metrics

Practical Notes



Metrics

Outline

9

Regression Metrics

Classification
Metrics

Which metric should we report?

- MAE: robust, easy to interpret.
- RMSE: emphasizes large errors, common in forecasting.
- R^2 : relative improvement over mean baseline.

Always report context

- Units and scale of y (consider normalization).
- Baseline: mean predictor or a simple linear model.
- Outliers and heavy-tailed noise can flip conclusions.



Metrics

Outline

Regression Metrics

10 Classification Metrics

Classification Metrics

Classification Metrics

Problem Setting and Confusion Matrix

Binary Classification

$y \in \{0, 1\}$, model outputs $\hat{p} = P(y = 1|\mathbf{x})$ and a label

$$\hat{y} = \mathbf{1}\{\hat{p} \geq \tau\}$$

Confusion Matrix (Binary)

	$\hat{y} = 1$	$\hat{y} = 0$
$y = 1$	<i>TP</i>	<i>FN</i>
$y = 0$	<i>FP</i>	<i>TN</i>

Key Idea

Most classification metrics are functions of (TP, FP, TN, FN) .



Metrics

Outline

Regression Metrics

11 Classification Metrics

Classification Metrics

Accuracy, Precision, Recall, F1



Core Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

When Accuracy is misleading

If classes are imbalanced, Accuracy may look high even for a weak model.
Prefer Precision/Recall/F1 and PR-AUC.

Metrics

Outline

Regression Metrics

12 Classification Metrics

Classification Metrics

Step-by-step Example: Confusion Matrix

Example (n=10)

Ground truth:

$$\mathbf{y} = [1, 0, 1, 1, 0, 1, 0, 0, 1, 0]$$

Predicted labels:

$$\hat{\mathbf{y}} = [1, 0, 0, 1, 0, 1, 1, 0, 1, 0]$$

Count outcomes (positive class = 1):

- TP : predict 1 when $y = 1$ at positions $\{1, 4, 6, 9\} \Rightarrow TP = 4$
- FN : predict 0 when $y = 1$ at position $\{3\} \Rightarrow FN = 1$
- FP : predict 1 when $y = 0$ at position $\{7\} \Rightarrow FP = 1$
- TN : remaining zeros correctly predicted $\Rightarrow TN = 4$

$$(TP, FP, TN, FN) = (4, 1, 4, 1)$$

Metrics

Outline

Regression Metrics

13 Classification Metrics

Classification Metrics

Step-by-step Example: Compute Metrics

Using $(TP, FP, TN, FN) = (4, 1, 4, 1)$:

$$\text{Accuracy} = \frac{4 + 4}{10} = 0.8$$

$$\text{Precision} = \frac{4}{4 + 1} = 0.8 \quad \text{Recall} = \frac{4}{4 + 1} = 0.8$$

$$F1 = 2 \cdot \frac{0.8 \cdot 0.8}{0.8 + 0.8} = 0.8$$

Interpretation

Here Precision = Recall = F1, but this is not always true. Trade-offs appear when FP and FN behave differently.



Metrics

Outline

Regression Metrics

14 Classification Metrics

Classification Metrics

Specificity and Balanced Accuracy

Specificity (True Negative Rate)

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Balanced Accuracy

$$\text{BalancedAcc} = \frac{1}{2} (\text{Recall} + \text{Specificity})$$

Why it matters

Balanced Accuracy is useful for imbalanced datasets because it treats both classes symmetrically.



Metrics

Outline

Regression Metrics

15 Classification Metrics

Classification Metrics

Thresholding: Precision–Recall Trade-off

Predicted probabilities $\hat{p}$ and threshold τ

$$\hat{y} = \mathbf{1}\{\hat{p} \geq \tau\}$$

Higher τ usually increases Precision but decreases Recall.

Mini Example (n=6)

$$\mathbf{y} = [1, 1, 1, 0, 0, 0], \quad \hat{\mathbf{p}} = [0.9, 0.6, 0.4, 0.7, 0.3, 0.2]$$

- If $\tau = 0.5$: $\hat{\mathbf{y}} = [1, 1, 0, 1, 0, 0] \Rightarrow TP = 2, FN = 1, FP = 1, TN = 2$
- If $\tau = 0.8$: $\hat{\mathbf{y}} = [1, 0, 0, 0, 0, 0] \Rightarrow TP = 1, FN = 2, FP = 0, TN = 3$



Metrics

Outline

Regression Metrics

16 Classification Metrics

Classification Metrics

Compute Metrics at Two Thresholds

From the previous mini example:

At $\tau = 0.5$ (TP=2, FN=1, FP=1, TN=2)

$$\text{Precision} = \frac{2}{2+1} = 0.667, \quad \text{Recall} = \frac{2}{2+1} = 0.667$$

At $\tau = 0.8$ (TP=1, FN=2, FP=0, TN=3)

$$\text{Precision} = \frac{1}{1+0} = 1.0, \quad \text{Recall} = \frac{1}{1+2} = 0.333$$

Takeaway

Choosing τ depends on the application costs of FP vs FN.

Metrics

Outline

Regression Metrics

17 Classification Metrics

Classification Metrics

Multi-class Metrics: Macro vs Micro

Multi-class Setting

$y \in \{1, \dots, K\}$, often evaluated by per-class Precision/Recall/F1.

Macro-average

Compute metric per class, then average:

$$\text{MacroF1} = \frac{1}{K} \sum_{k=1}^K \text{F1}_k$$

Micro-average

Aggregate TP/FP/FN over all classes, then compute:

$$\text{MicroF1} = \frac{2TP}{2TP + FP + FN}$$



Metrics

Outline

Regression Metrics

18 Classification Metrics

21

Classification Metrics

Step-by-step Multi-class Example ($K=3$)

Example ($n=8$), classes: A,B,C

True labels:

$$\mathbf{y} = [A, A, B, B, B, C, C, C]$$

Predicted:

$$\hat{\mathbf{y}} = [A, B, B, B, C, C, A, C]$$

True \ Pred	Pred		
	A	B	C
A	1	1	0
B	0	2	1
C	1	0	2

- For class A: $TP_A = 1$, $FP_A = 1$ (from C), $FN_A = 1$ (A predicted as B)



Metrics

Outline

Regression Metrics

19 Classification Metrics

Classification Metrics

Compute Macro Precision/Recall/F1 (Example)

From the previous confusion matrix:

Class A

$$P_A = \frac{1}{1+1} = 0.5, \quad R_A = \frac{1}{1+1} = 0.5, \quad F1_A = 0.5$$

Class B and C

$$P_B = \frac{2}{2+1} = 0.667, \quad R_B = \frac{2}{2+1} = 0.667, \quad F1_B = 0.667$$

$$P_C = \frac{2}{2+1} = 0.667, \quad R_C = \frac{2}{2+1} = 0.667, \quad F1_C = 0.667$$

$$\text{MacroF1} = \frac{0.5 + 0.667 + 0.667}{3} \approx 0.611$$

Metrics

Outline

Regression Metrics

20 Classification Metrics

21

Metrics Summary

What to report in practice

Regression

- Report: $\text{MAE} + \text{RMSE (or MSE)} + R^2$
- Always include baseline and error distribution checks

Classification

- Balanced dataset: $\text{Accuracy} + \text{F1}$
- Imbalanced dataset: $\text{Precision/Recall/F1} + \text{PR-AUC}$ (and confusion matrix)
- Always specify threshold τ (if using hard labels)

Instructor Note

Metrics are not just numbers: they encode the cost of mistakes.



Metrics

Outline

Regression Metrics

21 Classification Metrics